

Lesson 3

Material

<https://github.com/unixfile/learning-python-2026>

Today

- ▶ Lists: creation, indexing, methods
- ▶ Slicing
- ▶ Copying lists
- ▶ Multidimensional lists
- ▶ Mutability vs. immutability
- ▶ Tuples

1. Creating a list

```
In : fruits = ["banan", "melon", "kiwi", "citron"]
```

```
In : type(fruits)
```

```
Out: list
```

```
In : len(fruits)
```

```
Out: 4
```

2. Indexing

```
In : fruits = ["banan", "melon", "kiwi", "citron"]
```

```
In : fruits[0]
```

```
Out: 'banan'
```

```
In : fruits[-1]  # -1 is the last element
```

```
Out: 'citron'
```

3. Modifying elements

```
In : fruits = ["banan", "melon", "kiwi", "citron"]
```

```
In : fruits[0] = "apelsin"
```

```
In : fruits
```

```
Out: ['apelsin', 'melon', 'kiwi', 'citron']
```

4. List methods

```
In : fruits = ["banan", "melon", "kiwi", "citron"]
```

```
In : fruits.append("mango")
```

```
In : fruits.pop()
```

```
Out: 'mango'
```

```
In : fruits.insert(0, "vindruva")
```

```
In : fruits.remove("kiwi")
```

```
In : fruits
```

```
Out: ['vindruva', 'banan', 'melon', 'citron']
```

5. Sorting

```
In : nums = [3, 1, 4, 1, 5, 9, 2, 6]
```

```
In : nums.sort()
```

```
In : nums
```

```
Out: [1, 1, 2, 3, 4, 5, 6, 9]
```

```
In : sorted([3, 1, 4])
```

```
Out: [1, 3, 4]
```

- ▶ `sort()` modifies in place; `sorted()` returns a new list

6. Slicing

```
In : s = [0, 1, 2, 3, 4, 5]
```

```
In : s[1:4] # includes 1, excludes 4
```

```
Out: [1, 2, 3]
```

```
In : s[:3]
```

```
Out: [0, 1, 2]
```

```
In : s[3:]
```

```
Out: [3, 4, 5]
```

```
In : s[::2]
```

```
Out: [0, 2, 4]
```

```
In : s[::-1]
```

```
Out: [5, 4, 3, 2, 1, 0]
```

7. Copying lists

```
In : a = [1, 2, 3]
```

```
In : b = a
```

```
In : b.append(4)
```

```
In : a
```

```
Out: [1, 2, 3, 4]
```

```
In : c = a.copy()
```

```
In : c.append(5)
```

```
In : a
```

```
Out: [1, 2, 3, 4]
```

- Assignment copies the reference, not the list

8. Multidimensional lists

```
In : matrix = [[1, 2, 3],  
               :      [4, 5, 6],  
               :      [7, 8, 9]]  
In : matrix[1][2]  
Out: 6
```

9. Mutability

```
# Lists are mutable  
numbers = [1, 2, 3]  
numbers[0] = 99 # ok  
  
# Strings are immutable  
s = "hello"  
s[0] = "H" # TypeError
```

10. Tuples

```
In : t = (1, 2, 3)
```

```
In : type(t)
```

```
Out: tuple
```

```
In : t[0]
```

```
Out: 1
```

```
In : t[0] = 99 # TypeError: object does not support item a
```

11. Tuples, continued...

```
In : point = (10, 20)
```

```
In : x, y = point
```

```
In : x
```

```
Out: 10
```

```
In : single = (42,)
```

```
In : type(single)
```

```
Out: tuple
```

- ▶ Use tuples for fixed data, lists for dynamic data